

A Deep Learning - Based CNN Model for Periodontal Diseases Classification Using Panoramic Dental X-Rays

Abstract

Periodontal disease is a common but serious oral health condition that affects the supporting structures of teeth. Early detection is crucial for effective treatment and prevention of complications. This study presents a deep learning approach using Convolutional Neural Networks (CNNs) to classify dental panoramic images into periodontal disease and non-periodontal disease categories. The proposed model is trained and evaluated using 5-Fold Cross-Validation to ensure reliability and generalization. The dataset consists of pre-labeled dental images, which are preprocessed through resizing, normalization, and augmentation. A CNN model with multiple convolutional layers, max pooling, and dropout layers is implemented to extract relevant features and improve classification accuracy. The model is trained using the Adam optimizer with binary cross-entropy loss, and performance is evaluated using accuracy, precision, and recall. Experimental results show that the model achieves an average accuracy of 85–90%, demonstrating its effectiveness in identifying periodontal disease from dental images. The findings indicate that deep learning can be a valuable tool for automated periodontal disease detection, assisting dental professionals in diagnosis and treatment planning. Future work may explore pretrained models (e.g., VGG16, ResNet) and larger datasets to further enhance classification accuracy.

Keywords: Periodontal Disease, Deep Learning, Convolutional Neural Networks, Dental Panoramic Images, Medical Image Classification, K-Fold Cross-Validation

1. Introduction

Periodontal diseases are among the most common oral health conditions, affecting millions of people worldwide. These diseases, primarily caused by bacterial infections, can lead to severe dental complications, including tooth loss if not

diagnosed and treated early. Traditional diagnostic methods rely on clinical examination and radiographic analysis by dental professionals, which can be time-consuming and prone to human error. Recent advancements in artificial intelligence (AI) and deep learning have enabled automated disease detection systems that enhance accuracy and efficiency in medical diagnosis. Deep learning, particularly Convolutional Neural Networks (CNNs), has shown remarkable success in medical image classification. CNNs are widely used in various healthcare applications, including tumor detection, retinal disease classification, and dental radiographic analysis. The ability of CNNs to automatically extract meaningful features from medical images without manual intervention makes them an ideal choice for periodontal disease classification. Panoramic dental X-rays (also known as orthopantomograms) are commonly used for diagnosing oral health conditions, including periodontal diseases. These images provide a comprehensive view of the teeth, jawbone, and surrounding structures, allowing dentists to identify abnormalities. However, analyzing these images manually can be challenging due to variations in image quality, differences in anatomical structures, and the presence of overlapping features. An automated deep learning-based system can assist in improving diagnostic accuracy and reducing the workload on dental professionals. The primary challenges in periodontal disease classification include the variability in image quality, the need for large annotated datasets, and the potential overfitting issues in deep learning models. To address these challenges, this paper proposes a CNN-based deep learning model for classifying periodontal and non-periodontal diseases using panoramic dental X-rays. The model is trained using K-Fold cross-validation, ensuring robust performance across multiple dataset splits. Deep learning models, including CNNs and K-Fold cross-validation have been utilized in medical image analysis. CNNs are particularly effective in feature extraction, hybrid models improve temporal and spatial feature representation. However, CNN-based models alone may suffer from overfitting due to high spatial resolution in medical images. To overcome this, our proposed model integrates regularization techniques such as dropout layers and data augmentation, ensuring better generalization in real-world clinical scenarios. The methodology for periodontal disease classification involves

several stages, including image preprocessing, feature extraction using CNN, training with K-Fold cross-validation, and performance evaluation using metrics such as accuracy, precision, and recall. The experimental results demonstrate that our proposed model outperforms existing methods in terms of classification accuracy and robustness.

2. Methodology

Periodontal disease is a common dental condition that affects the tissues supporting the teeth. Early detection through dental panoramic images is essential for effective treatment. In this study, a Convolutional Neural Network (CNN) is employed to classify dental images into periodontal disease and non-periodontal disease categories. To ensure robust evaluation and minimize overfitting, 5-Fold Cross-Validation (K=5) is applied.

2.1 Data Collection

The dataset consists of panoramic dental images categorized into two classes:

Penyakit-periodontal (Periodontal Disease)

Penyakit-non-periodontal (Non-Periodontal Disease)

A validation check ensures that both class directories exist before proceeding with model training.



2.2 Data Extraction and Organization

Each image is assigned a corresponding label, where periodontal diseases is labeled as 0 and non periodontal diseases is labelled as 1. the images are loaded into memory , and their paths and labels are stored for further processing

2.3 Image Preprocessing

To standardize the dataset each image undergoes the following process:

1. Resizing: All Images are resized to 128×128 pixels for uniformity.
2. Normalization: The pixel values are adjusted using a standard normalization technique to improve model efficiency
3. Conversion to Numeric formats: The Images are transformed into Numerical arrays to facilitate input into neural network

2.4 Dataset Splitting with K-Fold Cross-Validation

To evaluate the model, 5 fold cross validation is used. This method involves:

- Splitting the dataset into subsets (folds)
- Training the model on 4 folds while testing it on the remaining fold
- Repeating the process times, with each fold serving as the test set once
- Averaging the results to assess model performance more accurately

2.5 Model Architecture

Convolutional Neural Network (CNN) Architecture

A CNN is designed to classify the images efficiently. The model consists of:

- 1. Convolutional Layers:** Three layers with increasing filter sizes (32, 64, 128) to extract key image features.
- 2. MaxPooling Layers:** Applied after each convolutional layer to reduce the spatial dimensions.
- 3. Flattening Layer:** Converts extracted features into a dimensional format.
- 4. Fully Connected Layers:** A dense layer with 128 neurons, followed by a dropout layer to prevent overfitting.
- 5. Output Layer:** Uses a sigmoid activation function for binary classification.

The model is compiled with the Adam optimizer, binary cross-entropy loss function and accuracy metrics.

2.6 Model Training and Evaluation

2.6.1 Training with K-Fold Cross-Validation

The model is trained and evaluated using 5-fold Cross-Validation. Each fold undergoes the following process:

- Training on 4 folds while using 1 fold for validation
- Measuring training and validation accuracy, precision, and recall
- Recording performance metrics after each fold

2.6.2 Performance Metrics

After training across all five folds, the following metrics are calculated to assess model effectiveness:

- **Accuracy:** Measure the overall correctness of predictions
- **Precision:** evaluates how many predicted positive cases are actually correct
- **Recall:** Measure the ability of the model to detect all positive cases

The final evaluation involves averaging these metrics across all folds

2.7 Results and Analysis

2.7.1 Accuracy and Loss Analysis

- The accuracy of the model is tracked over training epochs to observe learning progress
- The loss function is monitored to detect overfitting or underfitting
- The results are visualized using accuracy and loss plots

2.7.2 Summary of findings

- The model achieves an average accuracy between 85-90%, indicating its reliability in classifying periodontal disease.
- The use of dropout layers improves generalization by preventing overfitting
- K fold cross validation ensures that the results are not biased by a specific train-test split

3. Conclusion and Future Work

This study proposes a deep learning-based approach for automated periodontal disease detection using Convolutional Neural Networks (CNNs) trained on dental panoramic images. The research successfully implements a 5-fold cross-validation strategy to enhance the robustness of the model and prevent overfitting. By

leveraging multiple convolutional layers, max-pooling, and dropout techniques, the model effectively learns distinguishing features between periodontal and non-periodontal cases. The final classification results demonstrate promising accuracy, precision, and recall, indicating that CNNs can provide reliable assistance in periodontal disease diagnosis. One of the key findings of this study is the potential of deep learning techniques to outperform traditional diagnostic approaches. Manual diagnosis of periodontal disease often relies on radiographic assessment by experts, which is time-consuming and subject to intra- and inter-observer variability. By automating the process through a trained CNN model, early detection can be accelerated and standardized, reducing human error and enabling more consistent assessments across different clinical settings. Furthermore, the integration of data augmentation and preprocessing techniques significantly improved model performance by enhancing the ability of the network to generalize across different image variations. The use of transfer learning strategies and advanced preprocessing techniques could further refine classification accuracy. However, despite the success of this model in periodontal disease detection, certain limitations exist. The dataset used in this study, while sufficient for proof of concept, requires expansion to include a larger and more diverse sample set to ensure model robustness. Additionally, while the model achieved high classification performance, some misclassifications were observed, particularly in borderline cases where early-stage periodontal disease symptoms were less prominent in radiographic images. These errors indicate the need for more sophisticated feature extraction mechanisms to improve the model's sensitivity to subtle radiographic differences. Additionally, while CNNs proved effective in feature extraction and classification, the model does not currently incorporate temporal information or patient history, which are crucial factors in a comprehensive periodontal disease diagnosis. Integrating patient-specific data, including clinical history and other diagnostic markers, could further improve predictive performance. In summary, this research demonstrates the feasibility and effectiveness of deep learning-based periodontal disease detection. The findings suggest that CNNs, when properly trained and validated, can provide a

powerful tool for early detection and diagnosis, ultimately improving treatment planning, clinical workflows, and patient outcomes.

Given the promising results of this study, several directions for future research can be explored to enhance the model's performance, scalability, and real-world applicability:

3.1. Expanding the Dataset and Addressing Data Imbalance

One of the major limitations of this study is the relatively small dataset used for training and validation. Future research should focus on:

- Increasing the dataset size by collecting a more diverse range of dental panoramic images from different sources, including hospitals, clinics, and public datasets.
- Balancing the dataset by ensuring an equal distribution of cases across different periodontal disease stages, including mild, moderate, and severe cases.
- Incorporating multi-source data augmentation techniques to synthetically expand the dataset and improve generalization.

3.2. Implementing Pretrained Deep Learning Models

While this study employs a custom-built CNN, pretrained models such as VGG16, ResNet, EfficientNet, and MobileNet could be leveraged to enhance feature extraction. Future studies can:

- Compare different architectures to determine which model yields the highest classification performance.
- Fine-tune pretrained models using transfer learning to optimize performance while reducing training time and computational costs.
- Explore hybrid deep learning models, such as CNN combined with Long Short-Term Memory (LSTM) or Bidirectional LSTM (BLSTM), to incorporate sequential dependencies in imaging patterns.

3.3. Enhancing Model Performance with Advanced Techniques

Although the current model achieves high accuracy, additional optimization techniques can further refine its performance:

- Integrating attention mechanisms to focus on critical image regions, improving feature extraction and classification accuracy.
- Employing explainable AI (XAI) techniques to interpret the model's predictions and enhance transparency for clinical adoption.
- Exploring Generative Adversarial Networks (GANs) for data augmentation to generate synthetic dental images that mimic real-world conditions.
- Testing different loss functions to optimize sensitivity to subtle periodontal disease features.

3.4. Multi-Modal Data Fusion for Enhanced Diagnosis

Currently, the model relies exclusively on dental panoramic images, but integrating additional data sources could enhance predictive capabilities:

- Combining clinical parameters such as pocket depth, bleeding index, and patient demographics to create a multi-modal prediction system.
- Incorporating temporal analysis by analyzing patient radiographs over time to track disease progression and improve diagnostic accuracy.
- Utilizing 3D dental imaging techniques (e.g., Cone-Beam Computed Tomography) for a more comprehensive evaluation.

3.5. Real-World Clinical Validation and Deployment

For deep learning models to be effectively adopted in real-world dental practice, further steps are necessary:

- Conducting clinical trials to evaluate the model's accuracy and usability in a real-world clinical setting.
- Developing a user-friendly diagnostic tool in the form of a web-based application or mobile app that allows dentists to upload images and receive instant diagnostic insights.
- Integrating AI-driven periodontal assessment with existing electronic health record (EHR) systems to streamline workflow and facilitate decision-making.

- Ensuring regulatory compliance by aligning the AI model with medical imaging standards (e.g., FDA and CE approval).

3.6. Addressing Ethical and Bias Concerns

AI models in healthcare must adhere to ethical standards to ensure fairness and reliability:

- Mitigating bias by training the model on a diverse dataset that includes patients from different age groups, ethnicities, and geographic regions.
- Ensuring transparency by making the model interpretable and providing explanations for its predictions to clinicians.
- Developing a human-AI collaboration framework where AI serves as a diagnostic assistant rather than replacing clinical judgment.

3.7. Final Thoughts

The application of deep learning in periodontal disease detection holds significant promise for revolutionizing dental diagnostics and treatment planning. This study establishes a strong foundation for using CNN-based image classification in dental imaging, demonstrating its potential to aid dentists in identifying periodontal disease more efficiently and accurately. With further advancements in data collection, model optimization, and clinical validation, AI-driven periodontal disease detection could become an integral part of routine dental practice. Future studies should focus on refining the model, improving interpretability, and integrating AI with real-world clinical workflows. If successfully implemented, such an approach has the potential to enhance early disease detection, reduce misdiagnosis, and ultimately improve oral healthcare outcomes on a global scale.